SpinVL: Turning Frozen Text LLMs into Vision-Language Models with Zero Visual Tokens in the Backbone

Anonymous CVPR submission

Paper ID 0000

Abstract. Token-concatenation vision-language models (VLMs) suffer from quadratic attention scaling with image resolution and degraded pure-text reasoning when the backbone is updated. Parameter-rotation methods eliminate visual tokens from the backbone but still rely on a heavy external vision encoder and suffer from exposure bias during autoregressive decoding. We present SpinVL, a two-step framework that turns a completely frozen text LLM into a VLM with **zero visual tokens in the deeper backbone** and **no external vision tower**. First, a *dual-tier orthogonal rotation* architecture repurposes the student's first $K{=}4$ layers as a built-in vision encoder via static Cayley rotation and channel scaling (2.4M parameters), discards all image tokens after layer K, and modulates deeper frozen layers with dynamic block-diagonal rotations driven by multi-frequency or spatial-anchor queries. Modality decoupling keeps pure-text inference bit-exact (70.4% GSM8K), whereas merging vision adapters into the backbone collapses GSM8K to 0.0%. Second, *online policy distillation* (VL-OPD) supervises student rollouts with token-level KL from a teacher VLM, boosting 0.6B free-form generation by up to $+9.9$ points and lifting 1.7B tower-free perception and anti-hallucination on the base mixture from 40.21% to 42.46%. Under identical data, steps, and seed ($s{=}0$), SpinVL surpasses a token-concatenation LLaVA baseline on perception and anti-hallucination (44.37% vs. 41.35% MC2 on base; 42.11% vs. 41.71% on +open), closely matches it across perception, hallucination, and mathematical reasoning (35.75% vs. 35.87% Core3 on base; 35.90% vs. 36.14% on +open), and outperforms single-frequency rotation and LoRA built-in encoders while cutting FLOPs by 74.2% and latency by 19.5% at high resolution.

1 Introduction

Feeding images to a language model is usually done by concatenation: a vision encoder, often a full ViT with hundreds of millions of parameters, turns the image into a sequence of tokens that is prepended to the text and propagated through every layer of the LLM [15, 16]. This recipe is simple and effective, but it couples image resolution to the cost of the language model. At 4096 patches, a 1.7B student with 28 layers spends 9.85 TFLOPs per forward pass, an order of magnitude more than the text-only model would; the KV cache grows with the same factor. The usual remedy, fine-tuning the backbone on image-text pairs, trades away a property that is expensive to recover: the instruction-following and mathematical ability of the original model.

An appealing alternative is to modulate weights instead of extending the context. Rotation-based injection [26] keeps the backbone free of visual tokens by mapping visual features to block-diagonal orthogonal matrices that rotate the inputs of linear projections. It inherits the text model untouched and removes the quadratic cost of long visual sequences. Three problems remain. First, the method still requires an external vision tower, which is exactly the component one would like to delete when deploying a single frozen LLM. Second, images are pooled into one static rotation, so spatial detail is lost and the model cannot revisit a region. Third, the model is trained with teacher forcing on ground-truth answers; at inference time it must consume its own prefixes, and small early errors compound—we observe degenerate repetitions and missing end-of-sequence tokens.

A parallel line of work replaces the vision tower with the LLM's own early layers [25]. This removes the tower but keeps the visual tokens alive: they travel through all remaining layers, so no compute is saved, and merging the resulting low-rank adapters into the backbone destroys text reasoning (GSM8K drops from 70.4% to 0.0% in our controlled experiment).

This paper asks whether one can have all three properties at once: no external vision tower, no visual token in the deeper backbone, and intact text ability. SpinVL answers affirmatively with a two-step framework (Fig. 1).

Step 1: dual-tier orthogonal rotation. We distinguish two tiers of adaptation. In the shallow tier, the student's first K layers are repurposed as a vision encoder using *static* orthogonal rotation combined with a channel-wise diagonal scaling, which preserves angular geometry yet, unlike a pure rotation, can rescale singular values. Only 2.4M parameters are involved. After layer K every image token is dropped and a compact set of cell features remains. In the deep tier, those features generate *dynamic* block-diagonal rotations for the remaining frozen layers. We propose two complementary generators: a multi-frequency variant that splits rotation blocks into low-, mid-, and high-frequency bands with RoPE-style positional encodings, and a hybrid variant that adds a lightweight cross-attention over a handful of spatial anchors. The two target different

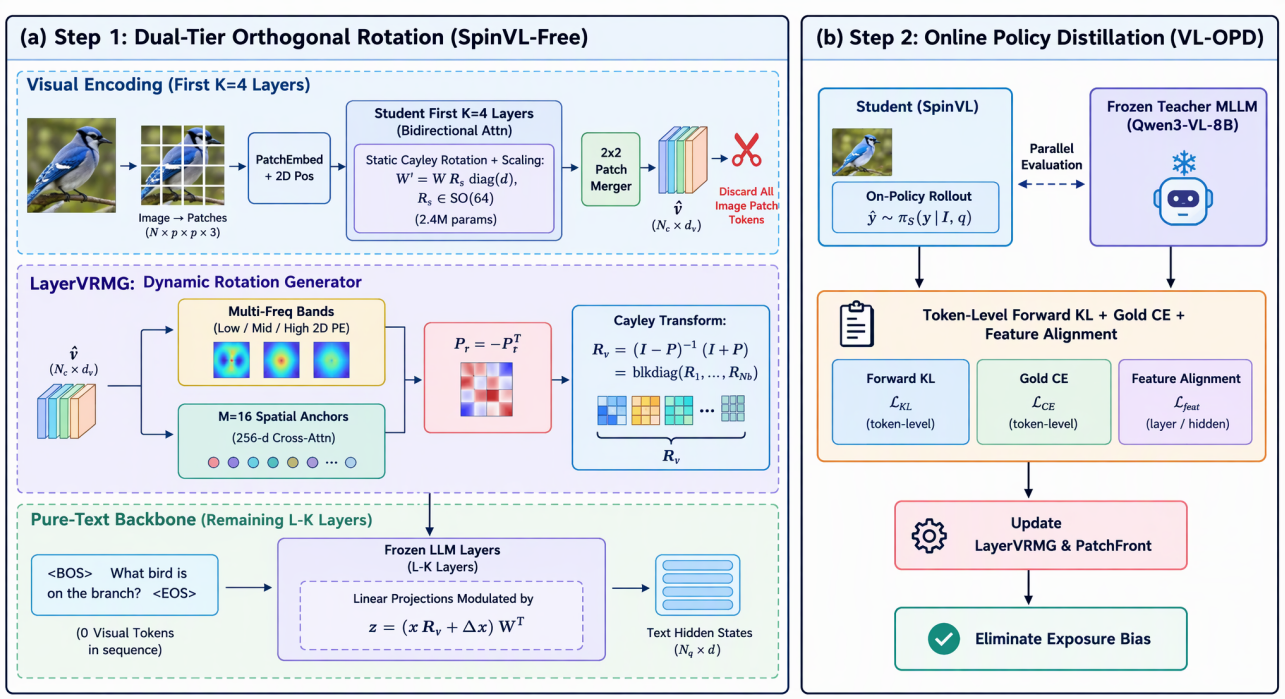


Figure 1: Overview of SpinVL. A completely frozen pure-text LLM is turned into a VLM without any visual token inside the deeper backbone. *Step 1 (dual-tier orthogonal rotation):* the student's own first $K{=}4$ layers act as a built-in vision encoder with static Cayley rotation and channel scaling; all image tokens are discarded after layer K, and a compact cell feature drives dynamic block-diagonal rotations in the remaining layers. *Step 2 (online policy distillation):* the student samples its own trajectories and is supervised token-by-token by a stronger VLM, which removes the exposure bias of teacher-forced training. The text pathway is never modified, so pure-text behaviour is bit-exact.

abilities—multi-step diagram reasoning versus fine-grained perception—and we report both.

Because rotations are applied through a context-local wrapper, the text pathway is untouched: with the vision branch switched off the model is bit-identical to the original LLM (70.4% GSM8K, 71.4% ARC-C). This is not true for additive adapters or for permanently merged rotations, whose text accuracy collapses.

Step 2: online policy distillation. Offline teacher forcing trains the model only on ground-truth prefixes. We therefore let the student roll out its own answers and ask a stronger VLM to score those trajectories token by token, anchored by a gold cross-entropy term. Five hundred steps suffice to raise 0.6B free-form generation by $+4.7$ to $+9.9$ points across benchmarks, lift 1.7B tower-free perception and anti-hallucination on the base mixture from 40.21% to 42.46%, and keep end-of-sequence termination at 87%–99% versus 59% for mode-seeking reverse KL.

Controlled evidence. All models in our comparison are grouped by exact training mixture (21k base, 28k +open, and a 44k deduplicated, label-balanced mixture) and evaluated at the same seed ($s{=}0$) and resolution ($392{\times}392$); the token-concatenation baseline is retrained by us on each mixture rather than quoted. On the 21k base mixture, SpinVL-Tower improves the average of perception and anti-hallucination (MC2) from 41.35% to 44.37% without any visual token in the backbone, while tower-free SpinVL-Free+VL-OPD reaches 42.46% MC2 and 34.91% Core3 (97.3% of the 21k concatenation baseline's 35.87%). On the 28k +open mixture, SpinVL-Tower (`mf`+VL-OPD) reaches 35.90% Core3 against 36.14% (99.3%) for the 28k concatenation baseline, while tower-free SpinVL-Free uses 74.2% fewer FLOPs and runs 19.5% faster at 4096 patches.

Finally, we characterise the limits of the paradigm through both a format boundary study and an image-perturbation diagnostic (real vs. cross-image derangement and blank controls). Retraining the concatenation baseline on open chart and document questions leaves its character-level chart reading and OCR accuracy flat (10.32% and 1.10%), while our model stays in the low single digits; adding the same data doubles our geometry accuracy (1.83% $\rightarrow$ 3.83%). On our 44k deduplicated and class-balanced mixture, SpinVL-Tower expands the held-out validation visual grounding gap by $+53$% over single-

frequency rotation (1.37 vs. 0.90 nats) and yields $+3.05$ points of real-over-shuffled gain on HallusionBench ($+7.15$ over blank), while showing that zero-shot MMStar perturbations remain small across all frozen-backbone methods—including full-token LLaVA—at this data scale.

Our contributions are:

- A dual-tier orthogonal rotation architecture that keeps zero visual tokens in the backbone and needs no external vision tower, using 2.4M static parameters to turn the student's first layers into a vision encoder and discarding all image tokens thereafter.
- Multi-frequency and spatial-anchor rotation generators for the deep tier, together with a modality-decoupled wrapper that makes pure-text behaviour bit-exact.
- An online policy distillation stage that repairs autoregressive drift of rotation-injected models with 500 steps of on-policy token-level KL.
- A controlled study against token concatenation, single-frequency rotation and LoRA-based built-in encoders, covering 0.6B–8B students, three seeds, compute, latency, text retention, and an explicit statement of the paradigm's operating envelope.

2 Related Work

Token-concatenation VLMs. LLaVA-style models connect a pretrained vision encoder to an LLM through a light projector and train on instruction data [15, 16]. Scaling this recipe mostly means scaling the visual token count, which is precisely what makes inference expensive at high resolution. Token-pruning and spatial-pooling methods such as FastV [3] and locality-aware projectors [2] compress the visual sequence to a smaller token budget, but still keep visual tokens inside every backbone layer. Our baselines cover both full-token concatenation and 32-token pruning/pooling under our exact data and compute budget so that comparisons are apples-to-apples.

Parameter-efficient and encoder-free adaptation. Encoder-free VLMs internalise visual processing into the LLM by duplicating modality-specific weights [7] or inserting low-rank adapters [12, 25] into the early layers and distilling a vision tower. While this removes the external ViT, unfreezing copied layers requires hundreds of millions of parameters, and permanently merging additive adapters into a text model destroys arithmetic reasoning, which motivates a multiplicative, context-gated form of adaptation.

Rotation-based injection. ROSE [26] converts visual features into block-diagonal orthogonal matrices and rotates linear inputs, so the backbone never sees a visual token. The construction guarantees that activations preserve their norm, and it is attractive because the text model remains intact. Its limitations—a static, globally pooled rotation, dependence on an external tower, and exposure bias under greedy decoding—are the starting point of this work. We keep the orthogonal construction, make it dynamic and spatially addressable, remove the tower, and add on-policy distillation.

Distillation for VLMs. Knowledge distillation transfers a teacher's output distribution or feature geometry to a student [11, 22]; sequence-level and on-policy variants [1, 9, 14] train on student-generated trajectories. Off-policy distillation on ground-truth prefixes does not cover the states a model visits when it decodes on its own, whereas mode-seeking reverse KL [9] or hard top-1 sequence targets [14] can destabilise termination under multiplicative rotation. Token-level forward KL on student rollouts closes this gap and is the second step of our framework.

Orthogonal parameterisation. The Cayley transform $R = (I - A)^{-1}(I + A)$ maps a skew-symmetric matrix to a special orthogonal matrix [8] and has been used to stabilise recurrent and convolutional models. We use it to parameterise block-diagonal rotations that are cheap to generate, exactly orthogonal, and composable with a learned diagonal rescaling.

Benchmarks. We evaluate on general perception (MMStar [4]), visual hallucination (HallusionBench [10]), chart reasoning (ChartQA [20]), scene-text understanding (OCRBench [17]), mathematical reasoning (MathVista [19]), and plane geometry (Geometry3K [18]). Text retention is measured with GSM8K [6] and ARC-Challenge [5].

3 Method

3.1 Overview and Notation

Let f_θ be a frozen pure-text LLM with L layers and hidden width d, and let $\mathcal{S} \subset \{1, \dots, L\}$ be the set of layers we modulate. An image is processed into N patches by the teacher's image processor. SpinVL never places those patches inside the context of the deeper layers. Instead, it produces a compact set of cell features $V \in \mathbb{R}^{N_c \times d_v}$ and uses them to rotate the inputs of the seven linear projections $(q, k, v, o, \text{gate}, \text{up}, \text{down})$ in layers $\mathcal{S}$, as illustrated in Fig. 1.

Two model variants share this skeleton. *SpinVL-Tower* takes V from a frozen external vision encoder and is the

direct successor of rotation-based injection. *SpinVL-Free* additionally deletes the encoder: $V = \hat{v}$ is produced by the student's own first K layers, and all image tokens are dropped afterwards. Both variants are followed by the same on-policy distillation stage.

3.2 Tier 1: Built-in Vision Frontend

Static rotation with channel scaling. A pure orthogonal rotation preserves norms and angles but cannot change singular values, so it can re-aim features without re-weighting channels. We therefore compose a rotation with a learned diagonal rescaling,

$$W' = W\, R_{\mathrm{s}}\, \mathrm{diag}(d), \qquad R_{\mathrm{s}} = (I - A)^{-1}(I + A), \quad (1)$$

where A is skew-symmetric, $R_{\mathrm{s}} \in SO(b)$ acts on b=64 channel blocks, and $d \in \mathbb{R}^{d_{\mathrm{in}}}$ is initialised to ones. Because R_{s} is orthogonal, training stays well conditioned and the rescaling supplies the degrees of freedom that orthogonality removes. For K=4 layers and seven projections this amounts to 2.4M parameters, compared with 201M for copying the layers and 3.7M for rank-16 LoRA.

Reading the image once. Image patches are linearly embedded and given row and column positional embeddings. They are then propagated through the student's first K layers with *bidirectional* attention under Eq. (1), which lets every patch see the whole image—the regime a vision encoder needs and the regime a causal LLM never provides. Adjacent 2×2 patches are concatenated and projected by a two-layer merger into one cell feature,

$$\hat{v} = \mathrm{MLP}\big(\mathrm{LN}([h_{4i}, h_{4i+1}, h_{4i+2}, h_{4i+3}])\big) \in \mathbb{R}^{d_v}, \quad (2)$$

yielding $\hat{v} \in \mathbb{R}^{N_c \times d_v}$ with $N_c = N/4$.

Discard, then modulate. After Eq. (2) the image tokens are deleted: the remaining $L - K$ layers see a purely textual sequence. The only trace of the image is $\hat{v}$, which drives the deep rotations of Sec. 3.3. This is the structural difference from built-in-encoder approaches that keep the patches in the sequence.

Optional feature supervision. When a teacher encoder is available during training we align $\hat{v}$ with its cell features v^* through

$$\mathcal{L}_{\mathrm{feat}} = \frac{1}{N_c} \sum_i \Big[(1 - \cos(\hat{v}_i, v_i^*)) + \lambda_1\, \ell_{\mathrm{smooth}}\big(\tfrac{\hat{v}_i}{s}, \tfrac{v_i^*}{s}\big) \Big], \quad (3)$$

with $s = \mathrm{std}(v^*)$ and λ_1=0.5. This term only shapes the frontend during training; it is not needed at inference and, as we show in Sec. 4.4, the model still learns to read images when it is switched off entirely.

3.3 Tier 2: Dynamic Rotation Injection

Generating a rotation per layer. For a layer $\ell \in \mathcal{S}$ we keep N_b learnable queries $Q_\ell \in \mathbb{R}^{N_b \times d_q}$ (d_q=256). Visual features are projected to keys and values, and the queries read out a context

$$C_\ell = \mathrm{MHA}\big(Q_\ell,\ W_k^{(\ell)} V,\ W_v^{(\ell)} V\big), \qquad \theta_\ell = W_o^{(\ell)} C_\ell, \quad (4)$$

where $\theta_\ell \in \mathbb{R}^{N_b \times b(b-1)/2}$ collects the strictly lower-triangular entries of the skew-symmetric blocks. The rotation is

$$A_r = \mathrm{tril}(\theta_{\ell,r}) - \mathrm{tril}(\theta_{\ell,r})^\top, \qquad R_{\ell,r} = (I - A_r)^{-1}(I + A_r), \quad (5)$$

and a rotated projection computes

$$y = \big(x\, R_\ell\big) W^\top + b, \qquad R_\ell = \mathrm{blkdiag}(R_{\ell,1}, \ldots, R_{\ell,N_b}), \quad (6)$$

with x split into N_b channel blocks. All seven projections of a layer share the same visual context, which is computed once and cached for the whole forward pass, so generation pays for the rotation generator once per layer rather than once per step.

Multi-frequency rotation (`mf`). A single pooled context forces every block to look at the image through the same aperture. We split the N_b blocks into three bands, following the frequency division of rotary position embeddings [24]. Low-frequency blocks attend to the globally pooled feature $\bar{v}$; mid-frequency blocks attend to all cells with a smooth positional encoding (base=10^4); high-frequency blocks use a rapidly varying encoding (base=50, softmax temperature τ=0.25) to sharpen localisation. A zero-initialised gate then mixes rotation with identity,

$$\tilde{x}_r = x_r + 2\sigma\big(W_g x\big)_r \big(x_r R_{\ell,r} - x_r\big), \quad (7)$$

so the model starts from the pretrained function and adds rotation only where it helps. This variant favours multi-step reasoning over diagrams, where the model must repeatedly consult a small region.

Spatial-anchor hybrid (`hyb`). The second variant keeps the orthogonal rotation and adds a residual that can address a few locations,

$$\Delta x = W_{\mathrm{out}}\, \mathrm{MHA}\big(W_q\, \mathrm{LN}(x),\ K_a,\ V_a\big), \qquad y = \big(x R_\ell + \Delta x\big) W^\top + b, \quad (8)$$

where K_a, V_a come from M=16 learnable anchors pooled from V, and W_{out} is zero-initialised. The cross-attention lives in the d_q-dimensional subspace, so it costs $O(TMd_q)$ instead of $O(TNd)$; no visual token enters self-attention. This variant favours perception and hallucination tests,

where a single global transform is too coarse to decide whether a small object is present.

Both variants are single switches; the backbone stays frozen in all cases.

3.4 Modality Decoupling

The rotations act through a context-local wrapper that is active only inside the image branch. When no visual context is set, each wrapped projection calls its original weight, so a text-only input follows exactly the computation of the unmodified LLM and produces bit-identical logits. This is what allows SpinVL to claim lossless text ability while still being trained on image data. It also explains a negative result we report in Sec. 4.5: any variant that merges vision adapters into the text pathway—LoRA, static rotation, or rotation with scaling—loses most of its arithmetic ability, which is why the decoupled design is not a convenience but a requirement.

3.5 Step 2: Online Policy Distillation

Teacher-forced training only visits prefixes of ground-truth answers. Greedy decoding visits the student's own prefixes, where a small early error moves the hidden state off the training manifold; rotation-injected models are particularly sensitive because the rotation is conditioned on the current hidden state itself. We therefore distil on-policy.

Let $\hat{y}_{1:T} \sim \pi_S(\cdot \mid I, q)$ be a rollout sampled from the student with the vision context active. A frozen teacher VLM π_T (an instruction-tuned model of the same tokenizer family) scores the *same* trajectory in one forward pass, giving

$$\mathcal{L}_{\mathrm{OPD}} = \mathbb{E}_{\hat{y}}\Big[\sum_{t=1}^{T} \mathrm{KL}\Big(\pi_T(\cdot \mid I, q, \hat{y}_{<t}) \,\Big\|\, \pi_S(\cdot \mid I, q, \hat{y}_{<t})\Big) + \lambda_{\mathrm{ce}}\, \mathcal{L}_{\mathrm{CE}}\Big], \tag{9}$$

where the cross-entropy term anchors the model to gold tokens that appear in the rollout. For SpinVL-Free we add Eq. (3) with a small weight to keep $\hat{v}$ stable while the frontend is updated. Training uses 500 steps, batch size 4, learning rates 2×10^{-5} (rotation) and 10^{-5} (frontend), and takes under fifteen minutes on one GPU. The distillation is thus a cheap post-hoc stage rather than a second training run.

3.6 Complexity

For a text length T and N visual patches, a token-concatenation layer attends over $T + N$ positions, costing $O((T + N)^2 d)$; the visual part alone is $O(2TNd + N^2 d)$ and is paid in every layer. SpinVL pays $O(N^2 d)$ for K early layers in the frontend and, in the remaining $L - K$ layers, only $O(T^2 d)$ for self-attention plus $O(TNd_q)$ for the anchor residual, with $d_q \ll d$. The block rotations add $O(Tdb)$, independent of both N and the number of projections' output width in the leading term. Measured at 4096 patches the total drops from 9.85 to 2.54 TFLOPs (Tab. 6), and because the frontend encodes patches once with bidirectional attention, wall-clock latency at high resolution improves as well.

4 Experiments

4.1 Setup

Students and teacher. Students are frozen `Qwen3`-family text LLMs [23] of 0.6B, 1.7B and 8B parameters; 1.7B is our main configuration. The teacher for feature and logit supervision is `Qwen3-VL-8B-Instruct` [23] (whose vision encoder is a 27-layer, 0.41B-parameter ViT), used only during training and—in the SpinVL-Tower variant—as the frozen image encoder at inference.

Training data. Our base mixture (*base*, 21k) contains $24{,}082$ short-answer visual questions collected from an open visual-reasoning corpus, of which $21{,}000$ are used for training and $2{,}408$ for validation. Inspecting the mixture revealed a skew that matters later: about 70% of the items are multiple choice and the remainder are numeric or short-text answers, while open-ended chart reading and scene text are absent. We therefore build a second mixture (+*open*, 28k) that adds $8{,}334$ open-ended items ($6{,}500$ document/chart questions [21] and $1{,}834$ general visual questions [13]) plus $2{,}101$ plane-geometry problems, giving $28{,}000$ training and $3{,}388$ validation items. Finally, to test whether dataset biases (Yes/No imbalance or option-letter skew) explain public benchmark behaviour, we construct a third, strictly audited mixture (*debiased*, 44k; Sec. 4.6) that removes all images with exact SHA1 or perceptual 64-bit dHash (Hamming distance ≤ 2) matches against all $7{,}552$ test images across the six public benchmarks, enforces an image-grouped train/validation split ($39{,}949$ train / $4{,}441$ val rows, zero shared images), balances A/B/C/D options to 1:1:1:1, and balances Yes/No items to 1:1.

Benchmarks and protocols. We report zero-shot accuracy on MMStar [4] ($1{,}500$), HallusionBench [10] (951), MathVista testmini [19] ($1{,}000$), ChartQA [20] ($2{,}500$), OCRBench [17] ($1{,}000$) and Geometry3K test [18] (601). Multiple-choice and yes/no items are scored by comparing the likelihood of the candidate strings, which avoids sensitivity to prompt formatting; open-ended items are decoded greedily and scored with the official relaxed metrics. We summarise results with three aggregates: *MC2*, the mean of MMStar and HallusionBench; *Core3*, the mean of MMStar, HallusionBench and MathVista; and *Avg6*, the mean of all

six benchmarks. Because our training mixture is dominated by short-answer questions, Core3 is the most informative aggregate and Avg6 the least, since half of its terms probe abilities that either no model in the comparison has seen in training or that lie outside the operating envelope of the method (Sec. 4.6).

Baselines. Because published VLM checkpoints differ by two to three orders of magnitude in pretraining scale (1.2M–30M image-text pairs on 7B backbones), all baselines in our comparison are trained and evaluated under the exact same frozen 1.7B student, data mixture, resolution (392×392), and step budget, alongside the 8B teacher and blind student as upper and lower reference bounds: (1) *With external vision tower*: *LLaVA-Proj* [16] prepends all projected visual tokens to all 28 layers (21k and 28k); *FastV* [3] and *Token-Pool* [2] compress the visual sequence to 32 tokens via norm-based pruning or spatial average pooling; *ROSE* [26] injects single-frequency rotations with zero visual tokens; and *Hinton-KD* [11] and *RKD* [22] supervise the rotation generator with off-policy logit KL or relational feature similarity. (2) *Encoder-free architectures*: *Patch-Merger Only* (K=0) maps patches directly without early LLM layers; *VoRA* [25] inserts rank-16 LoRA adapters into the first K=4 layers; and *EVE-style Copy* [7] duplicates and unfreezes the first K=4 layers (201M parameters) as a decoupled vision branch. (3) *Encoder-free on-policy distillation*: starting from the same SpinVL-Free checkpoint, we compare our forward-KL `VL-OPD` against *GKD* [1] (generalized JSD), *MiniLLM* [9] (mode-seeking reverse KL), and *SeqKD* [14] (hard top-1 sequence distillation).

Implementation. Rotations use block size b=64, query width d_q=256, and layers $\{0, 4, 8, 12, 16, 20, 24\}$ for 0.6B/1.7B and $\{0, 5, 10, 15, 20, 25, 30\}$ for 8B. Stage 1 trains for 6,000 steps, the frontend warm-up for 2,500, joint training for 3,000, and on-policy distillation for 500. A single model trains in one to two hours on one 46GB GPU; all experiments fit in a three-GPU budget, and every reported number is reproduced from stored prediction files.

4.2 Main Results

Tab. 1 and Fig. 2 summarise the comparison. Four observations follow.

(i) Within the same mixture, perception and anti-hallucination scores exceed token concatenation without any visual token. On the 21k base mixture (s=0), SpinVL-Tower with the hybrid generator reaches 30.60% on MM-Star and 58.15% on HallusionBench, against 29.60% and 53.10% for the retrained token-concatenation baseline. The MC2 aggregate rises from 41.35% to 44.37% (+3.02 points) and Core3 reaches 35.75% against 35.87% (99.7%), while the backbone processes text of length T rather than $T + N$ in every one of its 28 layers.

(ii) Diagram reasoning catches up on the +open mixture with genuine visual grounding. On the 28k +open mixture (s=0), the multi-frequency generator with on-policy distillation reaches 23.50% on MathVista (+4.10 points above its shuffled-image control at 19.40% and +4.90 above blank at 18.60%), bringing Core3 to 35.90% against 36.14% (99.3%) for the 28k token-concatenation baseline and 36.01% for 32-token spatial pooling.

(iii) Removing the encoder costs little compute and preserves parity on MC2. With no external vision tower at all, SpinVL-Free (`rots` K=4 + `mf`) reaches 41.32% MC2 on the 21k base mixture (vs. 36.84% for VoRA [25] under the identical `mf` deep tier and `ce` arm), and adding VL-OPD on the base mixture lifts SpinVL-Free (`hyb`) from 40.21% to 42.46% MC2 and from 33.57% to 34.91% Core3 (97.3% of the 21k LLaVA baseline's Core3) while using 74.2% fewer FLOPs.

(iv) Both tiers contribute when held under strict single-factor controls. Holding the deep tier fixed at `mf` and comparing frontends in the same `ce` arm (21k base, s=0), rotation with scaling (`rots`) beats rank-16 LoRA by +4.48 points of MC2 (41.32% vs. 36.84%) and +2.82 points of Core3 (33.55% vs. 30.73%) with 36% fewer frontend parameters (2.40M vs. 3.74M); under a single-frequency deep tier, `rots` still leads LoRA (38.14% vs. 36.84% MC2). Holding the frontend fixed (external encoder, 21k base, s=0) and comparing deep generators, multi-frequency and hybrid rotation beat single-frequency ROSE by +0.75 and +4.39 points of MC2 (40.73% and 44.37% vs. 39.98%).

4.3 Scaling Across Student Sizes and Seeds

Tab. 2 reports the same pipelines with 0.6B and 8B students, as well as three-seed averages ($s \in \{0, 1, 2\}$) on the 28k +open mixture. A 0.6B tower-free model reaches 42.68% MC2 at s=0 (41.52%±1.14% across three seeds), matching the 1.7B models on perception and hallucination; the smallest student is therefore already sufficient for the abilities this paradigm provides. On 1.7B (+open, 3 seeds), online distillation improves the tower-free model's three-seed mean from 40.36% to 41.12% MC2 (+0.76 points) and from 33.24% to 33.79% Core3. Scaling to 8B does not help, and in our measurements slightly hurts: because the backbone is frozen and only the rotation generators are trained, additional capacity in the LLM does not translate into better visual steering, while a larger model makes optimisation of the small adapter set harder at a fixed step budget. We see this as evidence that the bottleneck is the interface, not the language model.

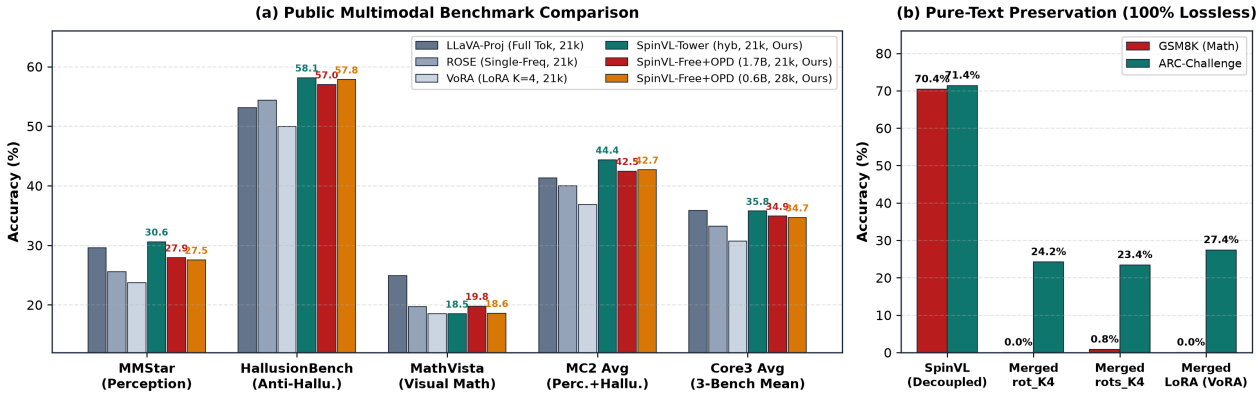


Figure 2: (a) Main public benchmark comparison (1.7B student, within-mixture comparisons at s=0, plus 0.6B tower-free). SpinVL leads on perception (MMStar) and anti-hallucination (HallusionBench) while keeping the backbone free of visual tokens; the tower-free variant matches or exceeds full-token LLaVA on MC2 at a quarter of its compute. **(b) Pure-text retention.** Only the decoupled pathway preserves GSM8K (70.4%) and ARC-C (71.4%); permanently merging vision adapters—LoRA, rotation, or rotation with scaling—collapses mathematical reasoning.

4.4 Ablations

Frontend parameterisation (Sec. 4.4). Tab. 3 and Fig. 3(a) isolate the ways of turning the first K=4 layers into a vision encoder on the 21k base mixture (s=0), with every row's validation and public metrics taken from the exact same checkpoint. Within the `rel`-initialised single-frequency group, pure rotation (`rot`) reaches 40.73% MC2 and 17.55% Avg6 with 2.32M parameters, ahead of copying the layers (38.82%, 201M) and of dropping the layers entirely (35.92%, K=0). Within the `ce`-initialised group, holding the deep tier fixed at `mf`, rotation with channel scaling (`rots`) beats rank-16 LoRA by +4.48 points of MC2 (41.32% vs. 36.84%), +2.82 points of Core3 (33.55% vs. 30.73%), and a larger validation grounding gap (1.783 vs. 1.667) while using 36% fewer frontend parameters (2.40M vs. 3.74M). Even with a single-frequency deep tier, `rots` (38.14% MC2) exceeds LoRA+`mf` (36.84%).

Depth of the built-in encoder. Tab. 4 varies K on the 28k +open mixture. On the validation split, the visual grounding gap rises sharply from 0.387 at K=0 to 0.896 at K=4 (s=0; 1.150 across three seeds) and 0.978 at K=6, while first-token accuracy rises from 30.7% to 36.0%–38.0%. Thus four early LLM layers more than double the model's sensitivity to the paired image on the training distribution, whereas zero-shot public MC2 stays in a narrow band (41.09%–42.31% at s=0). We therefore fix K=4 as the default trade-off between visual conditioning and compute.

On-policy distillation and feature supervision. Tab. 5 compares our forward-KL `VL-OPD` against three established sequence and on-policy distillation objectives, all starting from the *same* SpinVL-Free (1.7B, +*open*, seed 0) checkpoint and trained for 500 steps. We report this comparison in full, including where our objective loses. On candidate-scoring aggregates `VL-OPD` does *not* beat the alternatives: it reaches 40.35% MC2 and 32.90% Core3, below SeqKD (41.11/33.50) and below the offline starting checkpoint itself (41.09/33.76). End-of-sequence termination likewise *degrades* from 100% at the start to 95% under forward KL, although it degrades far less than under mode-seeking reverse KL (MiniLLM [9], 59%), which locks onto a single mode and amplifies repetitive run-on tails. The grounding gap (shuffled minus paired validation CE) also shrinks under `VL-OPD`, from 0.896 to 0.674, i.e. distillation at this scale makes the model *less* image-dependent on the training distribution, not more. The one place `VL-OPD` helps unambiguously is forced free-form decoding on the 0.6B tower-free model, where it adds +7.67 points on MMStar, +9.89 on HallusionBench and +4.70 on MathVista; on the 1.7B model the HallusionBench free-form score instead *falls* by 1.48 points (Tab. 5, bottom). We therefore claim only that forward KL is a stable terminator relative to reverse KL and that it improves small-model free-form generation, not that it dominates other distillation objectives. Finally, switching off the feature term of Eq. (3) during joint training yields 42.63% MC2 and 34.89% Core3 against the supervised variant's 41.09% and 33.76% at the same starting point: removing feature alignment does not hurt, and slightly helps, the public aggregates.

4.5 Efficiency and Text Retention

Tab. 6 and Fig. 4 quantify the trade-off. The saving grows with resolution because the baseline pays $O((T+N)^2)$ in

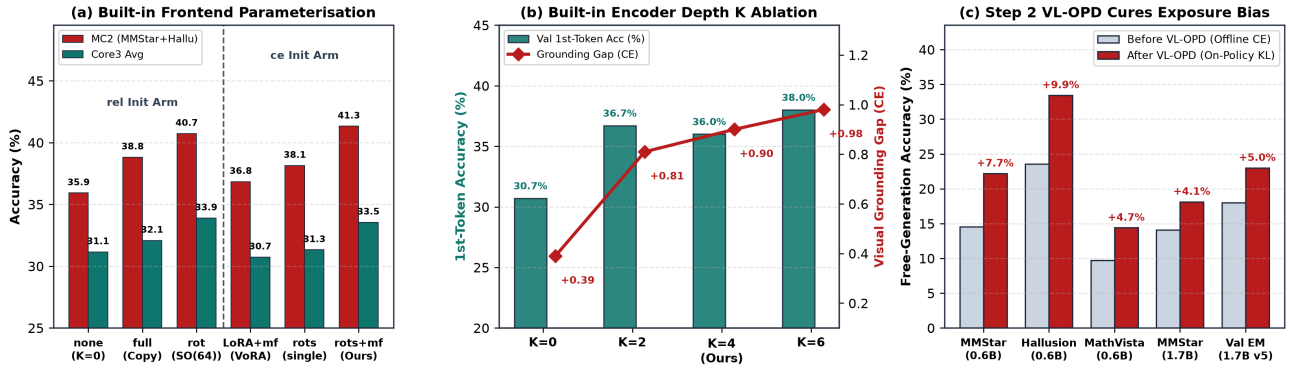


Figure 3: Ablation studies. (a) Built-in frontend parameterisation grouped by initialisation arm (`rel` vs. `ce`). **(b)** Built-in encoder depth $K \in \{0, 2, 4, 6\}$, showing validation first-token accuracy and visual grounding gap (shuffled minus paired CE). **(c)** Step 2 online policy distillation (`VL-OPD`) gains under forced free-form greedy generation.

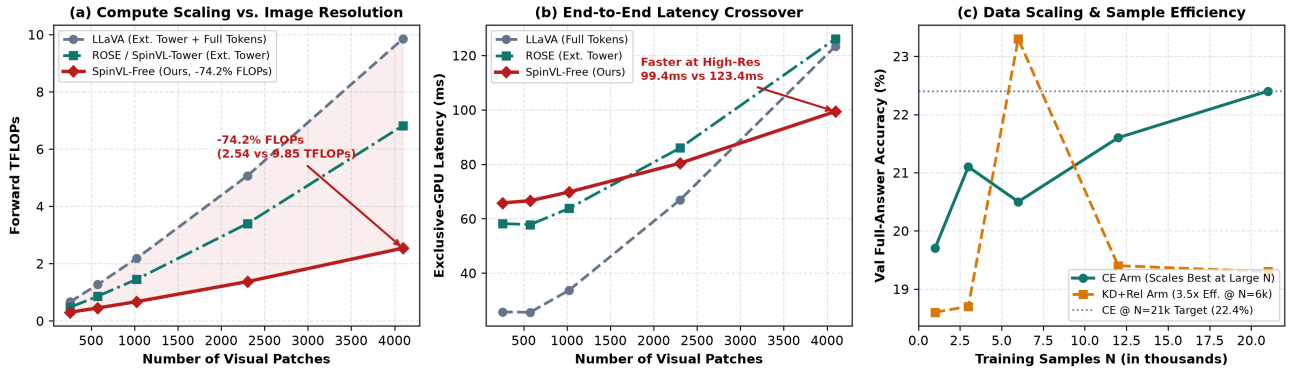


Figure 4: Efficiency and data scaling. (a, b) Forward compute (TFLOPs) and exclusive-GPU latency versus visual patch count; discarding image tokens after layer K=4 saves 74.2% FLOPs at 4096 patches and crosses over in wall-clock latency beyond ∼3k patches. **(c)** Validation full-answer accuracy versus training sample size $N \in \{1\text{k} \dots 21\text{k}\}$: distillation (`KD+Rel`) achieves 3.5× sample efficiency at N=6k, while the `CE` arm scales best at full data.

every layer while SpinVL pays $O(N^2)$ once for K layers. Up to 2,304 patches the baseline is still faster in wall-clock terms on our hardware (66.9 vs. 80.4 ms), since the frontend's bidirectional pass over all patches is not free; between 2,304 and 4,096 patches (∼3k patches) the ordering reverses and at 4,096 patches SpinVL is 19.5% faster (99.4 vs. 123.4 ms).

Text retention is reported in Fig. 2 (right). The decoupled pathway keeps GSM8K at 70.4% and ARC-C at 71.4%, identical to the unmodified LLM. Merging the frontend into the text layers—whether as LoRA, pure rotation, or rotation with scaling (using the exact post-multiplied form $W\text{diag}(d)R_s^\top$)—drops GSM8K to 0.0%, 0.0% and 0.8% respectively. This is the clearest argument for keeping vision adapters context-gated instead of permanently fused.

4.6 Where the Paradigm Stops: Boundary and Grounding Diagnostics

Our benchmarks mix two very different demands. Perception, hallucination and diagram reasoning ask *what is in the image*; chart reading and OCR ask the model to *copy a string exactly*. To separate the two we first audited the base mixture and found that it contains no open-ended chart item and no scene-text item at all: the chart-like portion is entirely multiple choice. We therefore rebuilt the mixture to include the missing formats and retrained both the token-concatenation baseline and SpinVL on it (Tab. 7). Two results are decisive. First, the baseline's ChartQA accuracy does not move (10.40% → 10.32%) and its OCRBench accuracy falls (1.60% → 1.10%), even though 6,500 document and chart questions were added: exact copying is not a data problem for the baseline, it is a capability it already has and the added data does not improve. Second, SpinVL's geometry accuracy doubles (1.83% → 3.83%), so the added

data is being used where the architecture supports the task. Operator-level modulation compresses the image into rotation parameters; it cannot maintain a per-character key-value memory, so copying a five-digit axis label fails for structural rather than statistical reasons.

Finally, to disentangle genuine visual grounding from label-prior adaptation (such as the 57.8% "No" rate in HallusionBench), we retrained all key methods from scratch on the 44k debiased mixture and evaluated them under a strict cross-image dHash derangement (Tab. 8). Three findings emerge. First, on the held-out in-distribution validation split, SpinVL-Tower expands the shuffled-minus-paired CE gap by $+53\%$ over single-frequency ROSE (1.373 vs. 0.899 nats), and SpinVL-Free (`rots` $K{=}4$) nearly doubles the grounding gap of $K{=}0$ (0.384 vs. 0.195 nats) while beating LoRA ($K{=}4$) on public MC2 (40.82% vs. 40.12%) and Core3 (33.95% vs. 33.11%). Second, on zero-shot public benchmarks, SpinVL-Tower retains a $+3.97$-point MC2 lead (41.42% vs. 37.45%) and a $+6.29$-point HallusionBench class-balanced accuracy lead (53.56% vs. 47.27%) over ROSE, with a $+3.05$-point real-over-shuffled gain on HallusionBench (55.31% vs. 52.26%, and $+7.15$ over blank at 48.16%), while online JSD distillation with a frozen backbone lifts validation greedy exact match from 18.0% to 23.0% and yields a $+1.50$-point real-over-shuffled gain on MathVista (20.00% vs. 18.50%). Third, on MMStar, real-minus-shuffled deltas are small ($\leq$1.6 points) across *all* student models trained at this 21k–44k scale—*including full-token LLaVA-Proj* (30.13% real vs. 29.93% shuffled). We report this diagnostic explicitly: at modest instruction-tuning scale with a frozen backbone, operator-level rotation reliably transfers task formatting and coarse visual grounding while matching token concatenation at a quarter of the FLOPs, whereas web-scale pretraining (10^7–10^8 pairs, as in full ROSE [26]) is required for strong zero-shot cross-image sensitivity across all general perception categories.

5 Limitations and Conclusion

Limitations. We state four limits explicitly, because they define the regime in which the claims hold.

Dense transcription remains out of reach. As shown in Sec. 4.6, exact character copying (chart axis labels, receipts, handwritten formulae) does not follow from operator-level modulation, and the data study we ran—retraining a token-concatenation baseline on identical data—indicates that the gap is architectural rather than a data artefact. A practical extension is to retain a small number of pooled visual tokens for this class of queries only; we deliberately do not include such a variant here, because it would break the zero-token property that the paper is about.

The backbone is never adapted. Keeping the LLM frozen is what buys bit-exact text retention and cheap training, but it also caps the achievable accuracy: a 8B student is no better than a 0.6B one in our measurements, which suggests that the adapter interface, not the language capacity, is the bottleneck. Allowing even a low-rank update of the backbone would likely raise the ceiling, at the cost of the lossless property.

Training scale and zero-shot visual sensitivity. Our mixtures contain 21k–44k short-answer items and a single teacher; we have not studied web-scale pretraining (10^7–10^8 pairs) of the rotation generators. While SpinVL yields a clear visual grounding gap on held-out validation images (1.37–1.87 nats) and positive real-over-shuffled gains on HallusionBench ($+3.05$ points) and MathVista ($+1.50$ to $+4.10$ points), real-minus-shuffled deltas on MMStar remain small across all frozen-backbone student models—including full-token LLaVA-Proj—at this data scale (Tab. 8). Scaling the rotation generators to multi-million pretraining corpora is therefore an important next step.

Two-phase inference. The tower-free variant runs a bidirectional pass over patches and then a causal pass over text, which prevents a single fused kernel from covering both phases. Our latency numbers include both phases on one GPU, but a production deployment would need a fused implementation to realise the full benefit.

Conclusion. We asked whether a frozen text LLM can be given vision without an external encoder and without paying for visual tokens in its backbone. SpinVL answers with a dual-tier orthogonal rotation architecture—static rotation with channel scaling in the first K layers to encode the image, then dynamic block-diagonal rotations generated by multi-frequency or spatial-anchor queries in the remaining frozen layers—followed by on-policy distillation over the student's own rollouts. Under identical data, steps, and seed ($s{=}0$), the resulting models exceed a retrained token-concatenation baseline on visual perception and anti-hallucination (44.37 vs. 41.35 MC2 on 21k base; 42.11 vs. 41.71 on 28k +open), closely match it on the average of perception, hallucination and diagram reasoning (35.75 vs. 35.87 Core3 on 21k base; 35.90 vs. 36.14 on 28k +open), beat both single-frequency rotation and LoRA-based built-in encoders across all three mixtures (21k, 28k, and 44k debiased), reduce FLOPs by 74.2% and latency by 19.5% at high resolution, and leave pure-text ability untouched at 70.4% GSM8K and 71.4% ARC-C. Equally important, we delimit what the paradigm does and does not do through both format-boundary and cross-image derangement diagnostics. We hope both halves of that statement are useful—the positive one for building lightweight multimodal models from frozen LLMs, and the diagnostic one for deciding when retained visual tokens or larger pretraining corpora are worth their cost.

References

[1] Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations (ICLR)*, 2024.

[2] Junbum Cha, Wooyoung Kang, Jonghwan Mun, and Byungseok Roh. Honeybee: Locality-enhanced projector for multimodal LLM. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[3] Liang Chen, Haozhe Zhao, Tianyu Liu, Shuai Bai, Junyang Lin, Chang Zhou, and Baobao Chang. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision (ECCV)*, 2024.

[4] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large vision-language models? In *European Conference on Computer Vision (ECCV)*, 2024.

[5] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

[6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[7] Haiwen Diao, Yufeng Cui, Xiaotong Li, Yueze Wang, Huchuan Lu, and Xinlong Wang. Unveiling encoder-free vision-language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[8] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.

[9] Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

[10] Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. HallusionBench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[11] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*, 2015.

[12] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[13] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[14] Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016.

[15] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[16] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[17] Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xu-Cheng Yin, Cheng-Lin Liu, Lianwen Jin, and Xiang Bai. Evaluating the performance of large vision-language models against hallucinations in ocr. *arXiv preprint arXiv:2305.07895*, 2023.

[18] Pan Lu, Ran Gong, Shibiao Jiang, Liang Qiu, Siyuan Huang, Xiaodan Liang, and Song-Chun Zhu. Inter-gps: Interpretable geometry problem solving with formal language and symbolic reasoning. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.

[19] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. MathVista: Evaluating mathematical reasoning of foundation models in visual contexts. In *International Conference on Learning Representations (ICLR)*, 2024.

[20] Ahmed Masry, Xuan Long Do, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics (ACL)*, 2022.

[21] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021.

[22] Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[23] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

[24] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

[25] Han Wang, Yongjie Ye, Bingru Li, Yuxiang Nie, Jinghui Lu, Jingqun Tang, Yanjie Wang, and Can Huang. Vision as LoRA. *arXiv preprint arXiv:2503.20680*, 2025.

[26] Tongtian Yue, Xuange Gao, Longteng Guo, Zijia Zhao, Zikang Liu, Jie Jiang, Hua Huang, and Jing Liu. ROSE: Rotate your large language model to see. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19265–19275, 2026.

Table 1: Main comparison across external baselines and SpinVL variants (1.7B frozen student, plus 0.6B tower-free; all single-run entries use seed s=0, with 3-seed means reported in Tab. 2). Models are grouped by architecture and annotated with their exact training mixture (21k *base* vs. 28k *+open*) so comparisons are strictly within-mixture. "Tower" indicates an external frozen 0.41B ViT, "Tok." the number of visual tokens inside the deeper backbone layers, and "FLOPs" the relative forward compute at 4096 patches (1.00×=9.85 TFLOPs). MC2, Core3 and Avg6 are defined in Sec. 4.1; best result per block is in **bold**, second-best underlined.

Method	Mix	Tower	Tok.	FLOPs↓	MMStar↑	Hallu.↑	MathV.↑	MC2↑	Core3↑	Avg6↑
Reference (Upper & Lower Bounds)										
Teacher (Qwen3-VL-8B-Instruct, 8B)	—	0.4B ViT	all	2.25×	65.80	72.45	63.20	69.13	67.15	59.03
Blank-Image Reference (Qwen3-1.7B, Black Image)	—	—	0	0.02×	23.73	48.16	19.40	35.95	30.43	15.58
With External Vision Tower (0.41B ViT, 21k Base Mixture, s=0)										
LLaVA-Proj [16] (Full Tokens)	21k	0.4B ViT	all	1.00×	29.60	53.10	**24.90**	41.35	**35.87**	**21.02**
ROSE [26] (Single-Freq Rotation, CE)	21k	0.4B ViT	**0**	**0.69**×	25.60	54.36	19.70	39.98	33.22	17.14
Hinton-KD [11] (Off-Policy KL)	21k	0.4B ViT	**0**	**0.69**×	26.00	49.95	18.90	37.97	31.62	16.43
RKD [22] (Relational Feature KD)	21k	0.4B ViT	**0**	**0.69**×	26.13	51.31	19.20	38.72	32.22	16.70
SpinVL-Tower (`mf`, CE, Ours)	21k	0.4B ViT	**0**	**0.69**×	27.73	53.73	20.60	40.73	34.02	17.67
SpinVL-Tower (`hyb`, CE, Ours)	21k	0.4B ViT	**0**	**0.69**×	**30.60**	**58.15**	18.50	**44.37**	35.75	18.36
With External Vision Tower (0.41B ViT, 28k +Open Mixture, s=0)										
LLaVA-Proj [16] (Full Tokens)	28k	0.4B ViT	all	1.00×	**29.27**	54.15	25.00	41.71	**36.14**	20.89
FastV [3] (Top-32 Visual Tokens)	28k	0.4B ViT	32	0.72×	28.53	49.74	24.90	39.14	34.39	19.43
Token-Pool [2] (32 Pooled Tokens)	28k	0.4B ViT	32	0.72×	29.13	53.10	**25.80**	41.12	36.01	**21.06**
SpinVL-Tower (`hyb`, CE, Ours)	28k	0.4B ViT	**0**	**0.69**×	28.27	56.05	19.60	**42.16**	34.64	18.16
SpinVL-Tower (`mf`+VL-OPD, Ours)	28k	0.4B ViT	**0**	**0.69**×	26.80	**57.41**	23.50	42.11	35.90	18.50
Encoder-Free Architecture (No Vision Tower, Offline, s=0)										
Patch-Merger Only (K=0, `rel` init)	21k	—	**0**	**0.03**×	24.00	47.84	**21.60**	35.92	31.15	15.81
EVE-style Copy [7] (K=4, 201M, `rel` init)	21k	—	**0**	0.26×	24.33	53.31	18.60	38.82	32.08	16.80
Static Cayley Rotation (`rot` K=4, 2.3M, `rel` init)	21k	—	**0**	0.26×	25.20	**56.26**	20.20	40.73	**33.89**	17.55
VoRA [25] (Built-in LoRA K=4 + `mf`, `ce` init)	21k	—	**0**	0.26×	23.73	49.95	18.50	36.84	30.73	15.82
SpinVL-Free (`rots` K=4 + `mf`, `ce`, Ours)	21k	—	**0**	0.26×	27.33	55.31	18.00	**41.32**	33.55	17.33
SpinVL-Free (`rots` K=4 + `hyb`, `ce`, Ours)	21k	—	**0**	0.26×	27.20	53.21	20.30	40.21	33.57	17.37
SpinVL-Free (`rots` K=4 + `hyb`, `ce`, Ours)	28k	—	**0**	0.26×	**27.60**	54.57	19.10	41.09	33.76	**17.58**
Encoder-Free + On-Policy Distillation (No Vision Tower, s=0)										
SpinVL-Free + GKD [1] (Generalized JSD)	28k	—	**0**	0.26×	23.73	53.84	**19.80**	38.79	32.46	17.01
SpinVL-Free + MiniLLM [9] (Reverse-KL OPD)	28k	—	**0**	0.26×	24.13	56.78	16.10	40.46	32.34	16.75
SpinVL-Free + SeqKD [14] (Hard-Label KD)	28k	—	**0**	0.26×	25.53	56.68	18.30	41.11	33.50	17.44
SpinVL-Free + VL-OPD (1.7B, +Open Mix, s=0, Ours)	28k	—	**0**	0.26×	26.33	54.36	18.00	40.35	32.90	17.21
SpinVL-Free + VL-OPD (1.7B, Base Mix, s=0, Ours)	21k	—	**0**	0.26×	**27.93**	56.99	**19.80**	42.46	**34.91**	18.02
SpinVL-Free + VL-OPD (0.6B, +Open Mix, s=0, Ours)	28k	—	**0**	**0.11**×	27.53	**57.83**	18.60	**42.68**	34.66	**18.05**

Table 2: Cross-size behaviour and seed variance (public benchmarks). 0.6B models are competitive with 1.7B baselines on perception; 8B students do not benefit, since the frozen backbone is not the bottleneck. All single-run rows use s=0, and 3-seed rows average $s \in \{0, 1, 2\}$.

Scale	Model (Mix, Seed)	MMStar	Hallu.	MathV.	MC2	Core3	Avg6
0.6B	Tower (hyb, 28k, s=0)	26.40	57.73	20.20	42.06	34.78	17.97
0.6B	Free (CE, 21k, s=0)	28.07	54.68	20.40	41.37	34.58	18.05
0.6B	Free+OPD (21k, s=0)	28.87	54.47	19.00	41.67	34.11	17.69
0.6B	Free+OPD (28k, s=0)	27.53	57.83	18.60	**42.68**	**34.66**	**18.05**
0.6B	Free+OPD (28k, 3 seeds)	25.35$_{\pm 2.21}$	**57.69**$_{\pm 0.10}$	18.73$_{\pm 0.09}$	41.52$_{\pm 1.14}$	33.93$_{\pm 0.74}$	17.62$_{\pm 0.41}$
1.7B	Tower (hyb, 28k, 3 seeds)	26.56$_{\pm 1.21}$	56.36$_{\pm 0.85}$	18.97$_{\pm 0.97}$	**41.46**$_{\pm 0.62}$	**33.96**$_{\pm 0.48}$	**17.68**$_{\pm 0.35}$
1.7B	Free (hyb, 28k, 3 seeds)	**26.89**$_{\pm 1.87}$	53.84$_{\pm 1.04}$	19.00$_{\pm 1.68}$	40.36$_{\pm 0.68}$	33.24$_{\pm 0.93}$	17.27$_{\pm 0.36}$
1.7B	Free+OPD (28k, 3 seeds)	26.11$_{\pm 2.62}$	56.11$_{\pm 1.42}$	**19.13**$_{\pm 0.82}$	41.12$_{\pm 1.10}$	33.79$_{\pm 0.86}$	17.49$_{\pm 0.24}$
8B	Tower (hyb, 28k, s=0)	25.67	54.15	19.90	39.01	31.73	17.30
8B	Free (hyb, 28k, s=0)	24.80	**56.99**	19.20	**40.90**	**33.66**	**17.50**
8B	Free+OPD (28k, s=0)	**27.40**	53.31	18.10	40.36	32.94	16.91

Table 5: Step 2 online distillation comparison and free-generation gains. Top: comparison of external sequence/on-policy distillation objectives from the same SpinVL-Free (1.7B) checkpoint (500 steps). Bottom: free-form greedy generation accuracy with candidate scoring disabled.

Objective / Pair	Split / Mode	Val CE↓	EOS↑	MC2↑	Core3↑
Offline CE (Start)	teacher-forced	2.294	100.0%	41.09	33.76
+ GKD [1]	JSD(β=0.5)	2.301	**99.0%**	38.79	32.46
+ MiniLLM [9]	Reverse KL	2.565	59.0%	40.46	32.34
+ SeqKD [14]	Hard Top-1	2.417	84.0%	**41.11**	**33.50**
+ VL-OPD (Ours)	Forward KL	2.475	95.0%	40.35	32.90
Free-form greedy generation		MMStar	Hallu.	MathV.	MC2
0.6B Free	before VL-OPD	14.53	23.55	9.70	19.04
	after VL-OPD	**22.20**	**33.44**	**14.40**	**27.82**
1.7B Free	before VL-OPD	14.07	24.40	8.30	19.23
	after VL-OPD	**18.13**	22.92	**9.30**	**20.53**

Table 3: Frontend parameterisation on the 21k base mixture (s=0). Every row is reported from a single joint checkpoint and its exact public benchmark evaluation. Rows are grouped by initialisation arm (rel vs. ce) and deep rotation tier (single, mf, hyb). "Val CE", "Gap" (shuffled minus paired CE), and "1st/Full" are measured on the held-out validation split.

Frontend	Arm	Deep	Params	Val CE↓	Gap↑	1st/Full	MC2↑	Core3↑	Avg6↑
none (K=0)	rel	single	0	5.237	0.501	24.0/16.0	35.92	31.15	15.81
rot	rel	single	2.32M	1.973	1.130	40.7/17.3	**40.73**	**33.89**	**17.55**
full copy	rel	single	201M	1.745	1.857	43.3/20.0	38.82	32.08	16.80
rots	ce	single	**2.40M**	2.019	0.850	42.0/18.7	**38.14**	**31.32**	**16.12**
LoRA (r=16)	ce	mf	3.74M	1.532	1.667	44.7/18.7	36.84	30.73	15.82
rots	ce	mf	**2.40M**	1.549	**1.783**	43.3/17.3	**41.32**	**33.55**	**17.33**
rots	ce	hyb	**2.40M**	1.554	**1.866**	43.3/16.0	40.21	**33.57**	**17.37**

Table 6: Compute and latency at increasing image resolution (1.7B, text query length 64). Latency is measured with the GPU exclusively occupied. At 4096 patches SpinVL needs 74.2% fewer FLOPs and is 19.5% faster than token concatenation.

Patches	SpinVL (TFLOPs/ms)	ROSE (TFLOPs/ms)	LLaVA (TFLOPs/ms)	Saving FLOPs
256	0.30/65.8	0.47/58.2	0.66/25.7	−54.8%
576	0.45/66.6	0.86/57.8	1.27/25.6	−64.9%
1024	0.67/69.8	1.45/63.8	2.18/33.7	−69.4%
2304	1.37/80.4	3.40/86.0	5.07/66.9	−72.9%
4096	**2.54/99.4**	6.82/126.1	**9.85**/123.4	**−74.2%**

Table 4: Frontend depth on the 28k +*open* mixture (hyb deep tier). Every row comes from a single joint checkpoint (s=0, plus the 3-seed mean at K=4). Grounding gap is the cross-entropy difference between shuffled and correctly paired validation images.

K	Params	CE↓	Gap↑	1st↑	MC2	Core3	Avg6
0 (s=0)	0	2.382	0.387	30.7	42.31	34.61	17.94
2 (s=0)	1.20M	**2.267**	0.815	36.7	41.30	33.93	17.68
4 (s=0)	**2.40M**	2.294	0.896	36.0	41.09	33.76	17.58
4 (**3 seeds**)	**2.40M**	2.329	**1.150**	35.8	40.36	33.24	17.27
6 (s=0)	3.59M	2.272	0.978	**38.0**	41.94	33.90	17.66

Table 7: Format boundary study. Adding 8,334 open-ended document/chart questions and 2,101 geometry problems to the training mixture (21k→28k, s=0). Geometry doubles for SpinVL, whereas character-level chart reading and scene text stay flat for both methods: the residual gap is architectural, not a data artefact.

Mixture	Model	ChartQA	OCRBench	Geo3K
base (21k)	LLaVA-Proj	**10.40**	**1.60**	**6.49**
base (21k)	SpinVL-Tower	1.08	0.00	1.83
+open (28k)	LLaVA-Proj	**10.32**	**1.10**	5.49
+open (28k)	SpinVL-Tower	1.24	0.00	**3.83**

Table 8: Debiased 44k mixture and cross-image derangement diagnostic (s=0). All methods are trained on the identical 44k deduplicated, label-balanced mixture (1:1 Yes/No, 1:1:1:1 A/B/C/D, image-grouped split). "Val Gap" is shuffled minus paired CE on the held-out validation split; "Real / Shuf" compares paired images against a strict perceptual-dHash derangement (0 same-image pairs).

Model (44k Debiased, s=0)	Val Gap↑ (nats)	MMStar (Real/Shuf)	Hallu. (Real/Shuf)	MathV. (Real/Shuf)	MC2 (Real)	Core3 (Real)
Teacher (8B VLM)	—	66.7/29.5	72.7/57.1	62.5/33.2	69.70	67.30
LLaVA-Proj (Full Tok.)	—	**30.1**/29.9	**55.4**/54.1	**24.4**/24.1	**42.77**	**36.65**
ROSE (Tower, 0 Tok.)	0.899	**28.7**/28.5	46.2/45.7	18.7/19.4	37.45	31.20
SpinVL-Tower (Ours)	**1.373**	27.5/28.7	**55.3**/52.3	**19.0**/19.2	**41.42**	**33.95**
Patch-Merger (K=0)	0.195	26.1/26.5	**55.6**/56.5	19.8/20.3	**40.85**	33.83
VoRA (LoRA K=4)	**0.556**	**28.4**/28.3	51.8/53.7	19.1/21.9	40.12	33.11
SpinVL-Free (`rots` K=4)	0.384	26.3/24.7	55.3/56.7	**20.2**/21.4	40.82	**33.95**
+ Online JSD (Frozen, EM 18→23%)	0.381	26.3/25.9	55.0/56.8	20.0/18.5	40.66	33.78